Lab 3: Component Modelling & Architectural Pattern Selection

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SECTION:D

Domain: Retail, E-Commerce & Finance
Stakeholders: Subscriber, Fulfillment Lead

- **Architecture Selection: "**We chose Microservices Architecture for the Customizable Subscription Box Scheduler System."

Two Reasons for Choosing Microservices Architecture

- Independent scaling: The Subscription Manager, Inventory Manager, and Fulfillment Manager may experience different workloads. Microservices allow each service to be scaled independently when the number of subscribers or scheduled boxes increases.
- Fault isolation and flexibility: The system has separate functions such as payment, box customization, inventory, and fulfillment. If one service temporarily fails, the other services can continue working, and individual services can be updated without affecting the entire system.

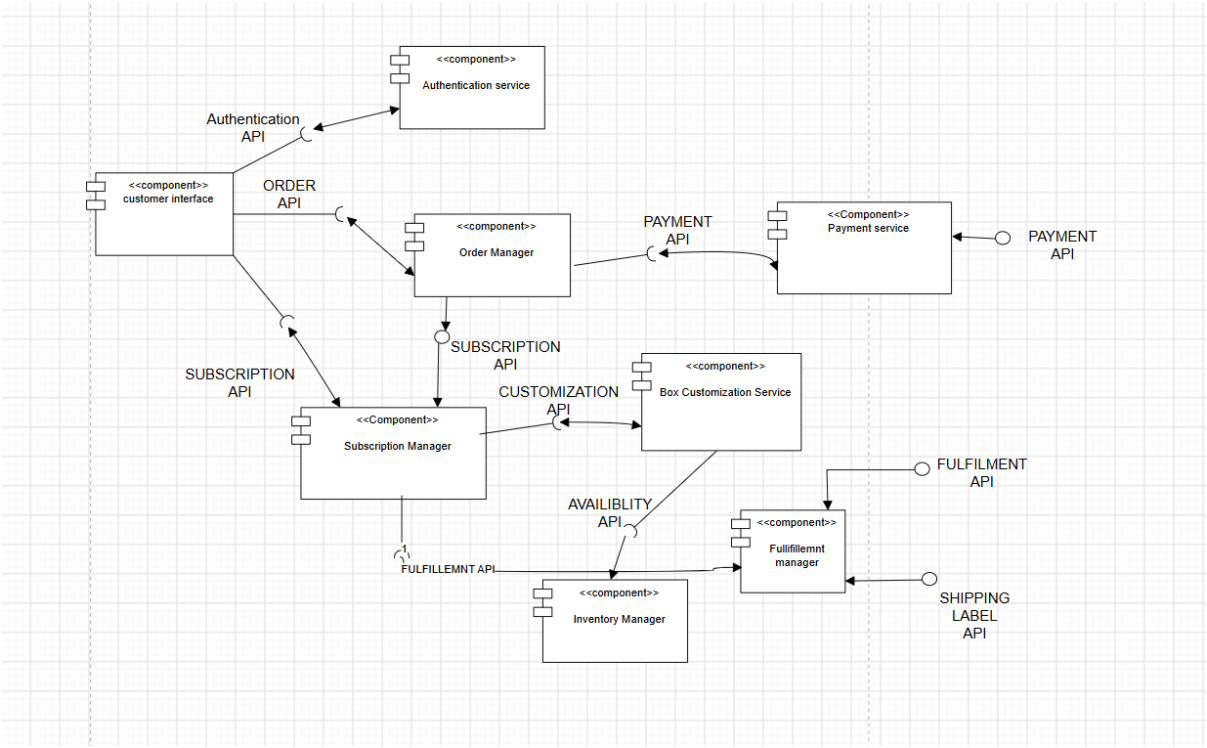
Security Advantage

Microservices Architecture provides better security by isolating different functions of the system into separate services. For example, the Payment Service handles sensitive payment information and can be protected using authentication, authorization, and secure API communication. Only authorized components, such as the Order Manager, can access the Payment Service. Similarly, customer information and subscription data can be protected within their respective services. This isolation reduces the impact of a security breach because unauthorized users cannot easily access the entire system through a single component.

Performance Benefit

Microservices Architecture improves the performance of the Customizable Subscription Box Scheduler by allowing different services to work independently and concurrently. For example, while the Order Manager processes a customer's subscription order, the Inventory Manager can check product availability and the Notification Service can prepare a notification without making the entire system wait for one large application process. Frequently used services can also be scaled independently when the number of subscribers increases. This helps the system handle more subscription requests efficiently and maintain good response time during high-demand periods.

DIAGRAM:



4. Component Diagram

The Component Diagram consists of eight major components:

1. Customer Interface

The Customer Interface provides the user-facing functionality through which subscribers can register/login, create subscriptions, customize their subscription boxes, place orders, and view order or subscription information.

2. Authentication Service

The Authentication Service manages customer authentication and authorization. It verifies user credentials and ensures that only authorized users can access subscription and order-related functions.

3. Order Manager

The Order Manager manages customer orders. It receives order requests, coordinates with the Subscription Manager, Payment Service, and Box Customization Service, and maintains the status of subscription orders.

4. Subscription Manager

The Subscription Manager manages subscription plans, subscription frequency, renewal information, and scheduled delivery dates. It communicates with the Fulfillment Manager to ensure that boxes are prepared according to the **customer's** schedule.

5. Box Customization Service

The Box Customization Service manages the products and preferences selected by the subscriber for their subscription box. It communicates with the Inventory Manager to verify whether the selected products are available.

6. Payment Service

The Payment Service processes subscription and order payments securely. It receives payment requests from the Order Manager and returns the payment status.

7. Inventory Manager

The Inventory Manager maintains product stock information. It checks product availability, reserves required products, and updates stock when products are allocated for subscription boxes.

8. Fulfillment Manager

The Fulfillment Manager manages the preparation and scheduling of subscription boxes for fulfillment and delivery. It coordinates with the Inventory Manager to ensure that required products are available.

No.	Interface	Connected Components	Purpose
1	Authentication API	Customer Interface ↔ Authentication Service	Handles login and authentication requests
2	Order API	Customer Interface ↔ Order Manager	Creates and retrieves customer orders
3	Subscription API	Order Manager ↔ Subscription Manager	Creates and manages subscriptions and schedules

4	**Payment API**	**Order Manager ↔ Payment Service**	**Processes subscription payments**
5	**Customization API**	**Order Manager ↔ Box Customization Service**	**Manages customer's box preferences**
6	**Inventory API**	**Box Customization Service ↔ Inventory Manager**	**Checks product availability**
7	**Fulfillment API**	**Subscription Manager ↔ Fulfillment Manager**	**Schedules subscription box fulfillment**
8	**Stock Update API**	**Fulfillment Manager ↔ Inventory Manager**	**Reserves and updates inventory**

Github link: https://github.com/keerthibpatil/SELAB